When Will a Borrower Default?

Survival Analysis in Credit Risk

Luke Munnelly

Should these be the same outcome?

- One borrower defaults after three months.

Hypothetical loans.

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- One borrower defaults after three months.
- Another defaults after two years.

Hypothetical loans.

Should these be the same outcome?

- One borrower defaults after three months.
- Another defaults after two years.

Both defaulted. The timing is different.

Hypothetical loans.

A third borrower has not defaulted yet

Observed for six months

No default so far

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What happens after month six is not observed.

A binary label needs a horizon

- A classifier can predict default within 12 months.

Training and evaluation labels need appropriate follow-up.

A binary label needs a horizon

- A classifier can predict default within 12 months.
- Six months without default does not establish a 12-month label.

Training and evaluation labels need appropriate follow-up.

Which 12-month labels are known?

Optional explanation

Observed history	Default by month 12?
Default at month 3	
Default at month 24	
No default through month 6	

Hypothetical loans; no earlier competing exit in these histories.

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Which 12-month labels are known?

Optional explanation

Observed history	Default by month 12?
Default at month 3	Yes
Default at month 24	No
No default through month 6	Unknown

Hypothetical loans; no earlier competing exit in these histories.

Why survival analysis?

- Use when events occur.

Machine-learning methods can also model survival outcomes.

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- Use when events occur.
- Accommodate right-censored follow-up under suitable assumptions.
- Here: Cox for interpretable relative default hazards.

Machine-learning methods can also model survival outcomes.

Choose the event and the clock

Origination $\longrightarrow$ first recorded default

Elapsed time since the loan was issued

Follow-up can end in different ways

ILLUSTRATIVE · Hypothetical loans; months since origination



C: right censoring — active when observation ends.

D: closure — a competing event ending default exposure.

The survival function

First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The survival function

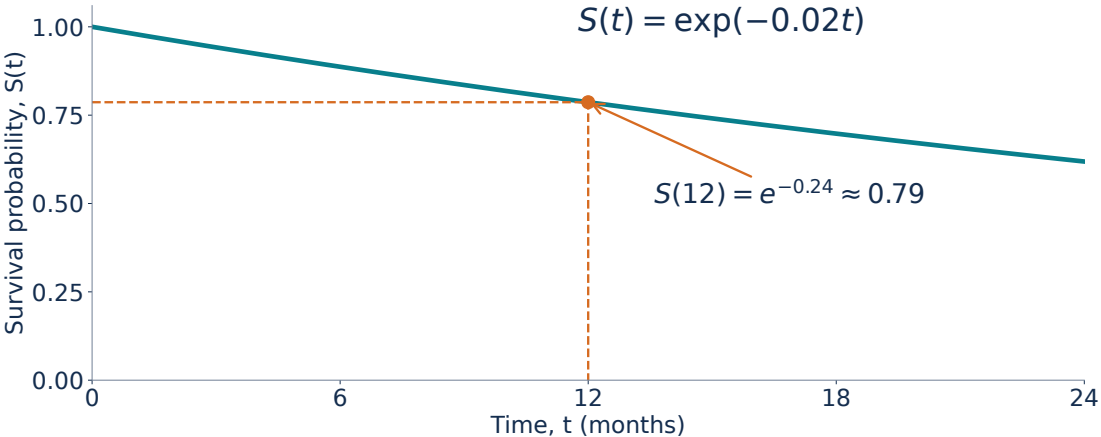
First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The probability of remaining default-free beyond time t.

Survival: the probability of remaining default-free

HYPOTHETICAL · Default-only setting; not a Bondora estimate



At 12 months, about 79% remain default-free in this hypothetical setting.

How rapidly are defaults occurring now?

Among loans still exposed now:

how rapidly do defaults occur next?

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Hazard: an instantaneous rate, not a probability.

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T_D < t + \Delta t \mid T_D \geq t)}{\Delta t}$$

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

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$$P(t \le T_D < t + \Delta t \mid T_D \ge t) \;\approx\; h(t)\,\Delta t$$

Rate $\times$ a short interval $\approx$ a conditional probability.

From hazard to survival

Optional explanation Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

From hazard to survival

Optional explanation Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

$$S(t) = \exp\{-H(t)\}$$

With competing closure, default hazard alone does not determine actual default probabilities.

Borrowers have different characteristics

Age Income Customer status

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Can these be associated with different default rates?

The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

For credit: a cause-specific default rate among loans still exposed.

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The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

Baseline: the time pattern for reference covariates.

Multiplier: how the covariates scale that rate.

For credit: a cause-specific default rate among loans still exposed.

Compare otherwise identical borrowers

Hypothetical new vs existing customer; same baseline and other covariates.

$$\mathrm{HR} = \frac{h(t \mid \text{new})}{h(t \mid \text{existing})} = \exp(\beta_{\text{new}})$$

A conditional rate ratio, not a probability ratio or causal effect.

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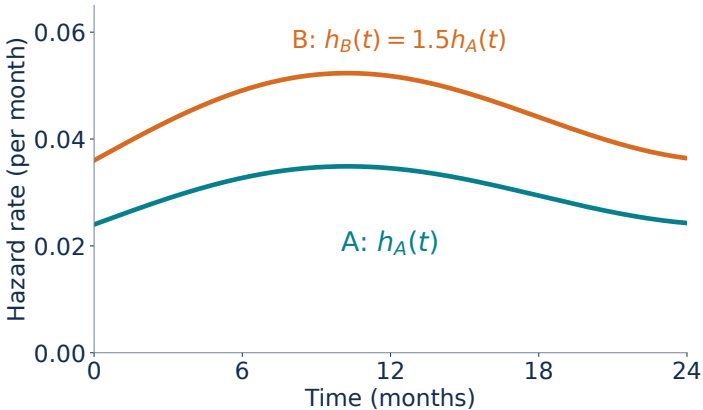
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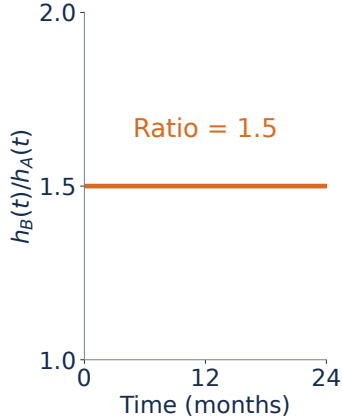
If $\mathrm{HR} = 1.5$: a 50% higher rate at each t, under PH.

A conditional rate ratio, not a probability ratio or causal effect.

Proportional hazards: changing rates, constant ratio

HYPOTHETICAL · A hazard is an instantaneous rate, not a probability





The baseline hazard can vary over time; proportionality concerns the ratio.

How does Cox learn relative effects?

Illustrative: three borrowers at risk just before one default.

A	B	C
Existing customer	Existing customer	New customer
Relative risk 1	Relative risk 1	Relative risk e^{β}

Which borrower accounts for this default?

Cox regression compares borrowers in the risk set

ILLUSTRATIVE · At risk immediately before a default; all other covariates equal

A	B	C
Existing customer	Existing customer	New customer
Relative risk: 1	Relative risk: 1	Relative risk: e^{β}

$$P(C \text{ defaults} \mid \text{one default in risk set}) = \frac{e^{\beta}}{2 + e^{\beta}}$$

If $\beta = \log(2)$: $\quad \frac{2}{1+1+2} = \frac{1}{2}$

Conditional on one default occurring now; not a probability over a time horizon.

The audited Bondora cohort

Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

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Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

9,844	6,924	2,508
Loans	Borrowers	Recorded defaults

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

Follow-up uses audited endpoints

- Default: first recorded `DefaultDate`.

Closure is not guaranteed full cash repayment. The source default threshold is unverified.

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Follow-up uses audited endpoints

- Default: first recorded `DefaultDate`.
- Closure proxy: `ContractEndDate` only when snapshot `Status=Repaid`.
- Stop at the first exit or 730 days.

Closure is not guaranteed full cash repayment. The source default threshold is unverified.

Five predictors and their contrasts

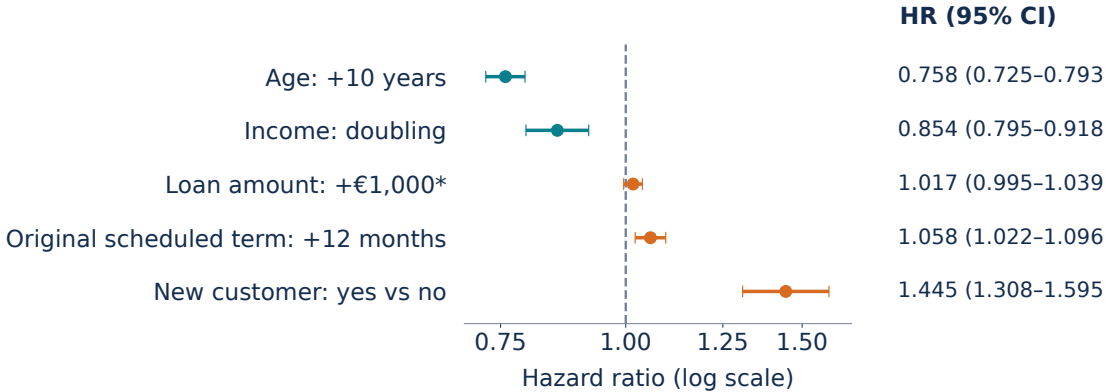
Age	+10 years
Income	Doubling
Amount	+1,000 recorded monetary units
Original scheduled term	+12 months
New-customer status	Yes vs no

Year-stratified baseline; borrower-clustered intervals.

Monthly original-schedule proxy; source currency/income period unverified.

Recorded default: adjusted hazard ratios

EMPIRICAL · Final audited Cox estimates; baseline stratified by origination year



Borrower-clustered 95% CIs are approximate with tied daily event times.

PH departures: age and new-customer status. Nonlinearity: age; borderline for term.

*Monetary fields as recorded; source currency/period definitions remain unverified.

Conditional associations, not causal effects or default probabilities. See REPORT.md.

New versus existing customer

$$\text{HR} = 1.445$$

95% CI: 1.308–1.595

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95% CI: 1.308–1.595

About 44% higher conditional default rate among loans still exposed.

Association, not a default-probability ratio or causal effect.

PH concern: this single HR is a model summary.

Where interpretation is limited

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- PH concerns: age and new-customer status.
- Nonlinearity; approximate tied-date intervals.
- Recorded endpoints and measurement proxies.

Back to the lending question

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- Three months and two years have different timing information.
- Six default-free months are useful, incomplete follow-up.
- Cox links covariates to conditional default rates.

Questions

Backup: partial likelihood

$$L(\beta) = \prod_{j:\,\text{default}} \frac{\exp(\beta^{\mathsf{T}} x_{i_j})}{\sum_{k \in \mathcal{R}_{y_j}(t_j)} \exp(\beta^{\mathsf{T}} x_k)}$$

$\mathcal{R}_{y_j}(t_j)$: loans still exposed immediately before t_j, within the defaulting loan's origination-year stratum.

Schematic formula for distinct event times. The audited fit uses Efron handling of ties.

Backup: corrected diagnostics

	PH rank p	PH log p	Nonlinear p
Age	0.00641	0.00768	0.00358
Log income	0.933	0.621	0.185
Amount	0.236	0.118	0.805
Original term	0.777	0.997	0.0438
New customer	0.00888	0.000417	—

PH: approximate independence-based Schoenfeld tests, not cluster-adjusted. Rank uses elapsed event times. Nonlinearity: existing spline cluster-Wald checks; exploratory.

One loan per borrower: 6,924 loans, 1,842 defaults. Broadly similar coefficient directions; sample composition changes.